Stablecoin Peg Stability and Risk-Aware Capital Allocation in DeFi

Engin Samet Dede
University of Geneva
Geneva, Switzerland
GitHub: https://github.com/Engn21

***Abstract*—While generally considered interchangeable dollar-denominated assets in decentralized finance (DeFi), their peg mechanism is defined by various factors, including the type of underlying collateral, governance structure, liquidity, and redemption mechanisms. Based on the daily price and trading-volume data of four major stablecoins (USDT, USDC, DAI, FRAX) obtained from the CoinGecko public API within the last 365 days, this paper analyzes the stability of these coins. The analysis is conducted based on absolute peg deviation, de-peg frequency, rolling volatility, volume-weighted deviation, and correlation with the overall stress in the crypto market. The results indicate that the risk of stablecoins is not uniform: the fiat-collateralized stablecoins have the smallest deviation on a day-to-day basis but a more centralized counterparty and custody risk; the on-chain risk profile of DAI is more apparent; and FRAX has a larger deviation behavior that is associated with its hybrid structure. The accompanying notebook reports a Kruskal-Wallis test that suggests that the distributions of deviations are statistically different for the four assets. The results are consistent with the principles of portfolio policies that would break down the allocation of stablecoins by mechanism and regularly check for the deviation from the peg, and not all stablecoins would be considered as one dollar coins.**

***Index Terms*—Stablecoins, DeFi, peg stability, market risk, capital allocation, non-parametric testing.**

I. Introduction

Stablecoins are an essential settlement and collateral mechanism in DeFi. They are applied in lending markets, automated market makers, derivatives collateral, treasury management and yield strategies [1]. They may appear simple at face value, but there are significant variations in the way the mechanisms are designed and how they fail. A portfolio that considers USDT, USDC, DAI and FRAX as all "cash-like" assets can therefore cause both the continuous peg noise and insufficient tail risk.

The study investigates the following question: how well do the major stablecoins maintain their \$1 peg, when do they fail to do so, and what does it mean for the risk-aware allocation of capital in DeFi? The contribution is an empirical workflow that builds comparisons between various stablecoin mechanisms with stablecoin deviation metrics and stablecoin visual diagnostics. The analysis focuses on average behavior and stress behavior because capital allocation is impacted not only by the average day tracking error but also by stress behavior's properties of liquidity, recovery time and market correlation.

TABLE I
Stablecoins Included in the Study

Asset	Ticker	Mechanism
Tether	USDT	Fiat-collateralized, centralized
USD Coin	USDC	Fiat-collateralized, centralized
Dai	DAI	Crypto-collateralized, MakerDAO
Frax	FRAX	Fractional-algorithmic, hybrid

II. Related Work

In stablecoin research, the assets are usually split based on their economic architecture that underpins the peg. Bullmann et al. [2] suggest a taxonomy of stablecoins according to issuer accountability, decentralization of responsibilities and asset backing. In Klages-Mundt et al. [3], the risks of stablecoins are summarized for both custodial and non-custodial designs. However, Lyons and Viswanath-Natraj demonstrate that price stability of stablecoins is closely tied to the nature of arbitrage, liquidity and the forces of demand for stablecoins, rather than just issuance [4]. Kozhan and Viswanath-Natraj examine DAI specifically and connect deviations from the peg of the stablecoin to the Collateral, Arbitrage friction and CDP activity [5].

Redemption rights, reserve quality, run risk, financial-stability spillovers are all highlighted in the regulatory literature. Gorton and Zhang also liken stablecoins to private bank money, and contend that the liabilities of the issuer can be run-prone if par convertibility is not guaranteed [6]. Other sources, such as the Financial Stability Board, the BIS, the IMF, and the U.S. President's Working Group, have noted vulnerabilities specific to stablecoins, including around their reserve assets, their ability to be redeemed and their operational risks, and the extent of their cross-border supervision [7]–[10]. This paper bridges that literature with an empirical, reproducible research workflow to measure peg behavior under various stablecoin mechanisms.

III. Assets and Mechanisms

The study covers investigation of the four stablecoins selected to represent distinct collateralization mechanisms. Table I summarizes the sample.

The comparison is intentionally mechanism-aware. While having limited observed price variation, Fiat-collateralized stablecoins also come with concentrated risks from issuer, custodian, regulatory, and redemption [10], [11]. Crypto-collateralized designs like DAI reveal more of the collateral and liquidation on-chain, but are subject to the volatility of the crypto-assets they are backing and the parameters of their governance rules [5]. Fractional or hybrid designs introduce endogenous risk due to a fall in the value of the endogenous collateral or governance token during the times when withdrawals are high [12].

IV. Data and Methodology

The daily close price and trading volume data is fetched from the CoinGecko endpoint `/coins/{id}/market_chart` [13] and is used to create daily datasets.Daily datasets are created from the CoinGecko endpoint `/coins/{id}/market_chart` which provides daily close prices and trading volumes. The notebook uses the data corresponding to the last 365 days at run time; thus, the time period used to create the sample will vary when the analysis is rerun. All prices are in U.S. dollars and are cleaned by sorting, de-duplicating dates, trimming to most recent 365 observations, and forward-filling isolated missing observations.

For stablecoin i on date t, the absolute peg deviation is defined as

$$d_{i,t} = |p_{i,t} - 1|, \tag{1}$$

where $p_{i,t}$ is the observed USD price. Two operational thresholds are used: a tight-peg band at $\pm 0.5\%$ and a de-peg band at $\pm 1.0\%$. A de-peg event is any day for which $d_{i,t} > 0.01$, with direction classified as a premium when $p_{i,t} > 1$ and a discount when $p_{i,t} < 1$.

The study computes the following risk metrics for each asset:

- mean, median, and maximum absolute deviation;
- standard deviation of daily absolute deviation;
- number of days outside the $\pm 0.5\%$ and $\pm 1.0\%$ bands;
- event recovery time, measured as consecutive days outside the $\pm 1.0\%$ band;
- 30-day rolling volatility of absolute deviation;
- 30-day volume-weighted peg deviation;
- Pearson correlation between absolute BTC daily returns and stablecoin deviation.

The volume-weighted deviation for stablecoin i over rolling window W_t is

$$\text{VWDev}_{i,t} = \frac{\sum_{\tau \in W_t} d_{i,\tau} v_{i,\tau}}{\sum_{\tau \in W_t} v_{i,\tau}}, \tag{2}$$

where $v_{i,\tau}$ is the trading volume reported for trader i on trading day τ. This ensures that low volume outliers are not treated in the same manner as high volume outliers.

Finally, a Kruskal-Wallis test is used to evaluate whether the four deviation distributions are statistically identical. Pairwise Mann Whitney U analysis is then used as a post-hoc analysis. These non-parametric tests are suitable because peg-deviations are not expected to follow a Gaussian distribution and it is a heavy-tailed distribution.

V. Empirical Results

Fig. 1 shows daily prices relative to the \$1 peg with the $\pm 0.5\%$ and $\pm 1.0\%$ bands. The visual result is consistent with the main thesis: most observations are close to \$1, but the assets do not share the same dispersion and stress behavior.

The spread distributions are compared in Fig. 2. The assets that are fiat-collateralized demonstrate the highest "peg" level of behavior day to day, whereas DAI and FRAX exhibit wider distributions. Based on the Kruskal-Wallis test, the accompanying notebook reports that the null hypothesis of equality of the 4 distributions is rejected at the 5% level of significance; thus, each stablecoin should be treated separately in portfolio risk models.

In Fig. 3 it is seen that average deviations from the monthly average are concentrated in time, meaning that peg risk clusters through time. This is of importance for allocation since it is common to base the allocation risk budget solely on the full-samples averages, which might become outdated in the changing market conditions.

Fig. 4 aggregates normalized risk dimensions into a relative profile. The radar chart is not a substitute for a formal risk model, but it is useful for quickly comparing how each asset ranks across deviation magnitude, volatility, and threshold breaches.

VI. Time-Varying and Liquidity-Adjusted Risk

Changes in the risk regime might be concealed in static summary statistics. The 30-day rolling volatility of peg deviation is shown in Fig. 5. High rolling volatility periods highlight that the risk of peg should be tracked over time, as it varies. For example, a rising rolling volatility can be an early warning signal of increasing market stress, liquidity issues, or emerging vulnerabilities in the peg mechanism. This can justify preemptive risk management actions such as reducing exposure, increasing collateral requirements, or rebalancing toward more liquid assets.

Fig. 6 reports 30-day volume-weighted deviation. This liquidity-adjusted measure is important for capital deployment because a deviation on thin volume has different implications from a deviation during heavy trading activity. Volume-weighting therefore helps identify deviations that are more likely to matter for large positions, liquidations, and redemption pressure.

VII. Market Stress Correlation

Broad crypto-market stress is proxied by the absolute daily BTC return. This follows prior evidence that stablecoin premiums, discounts, issuance, and arbitrage can interact with broader crypto-market liquidity and stress [4], [14]. Fig. 7 overlays normalized BTC price behavior with stablecoin peg deviations. The notebook also computes Pearson correlations between $|\Delta BTC|$ and each stablecoin's deviation. This diagnostic tests whether peg stress behaves as an isolated asset-specific event or clusters with broader market volatility.

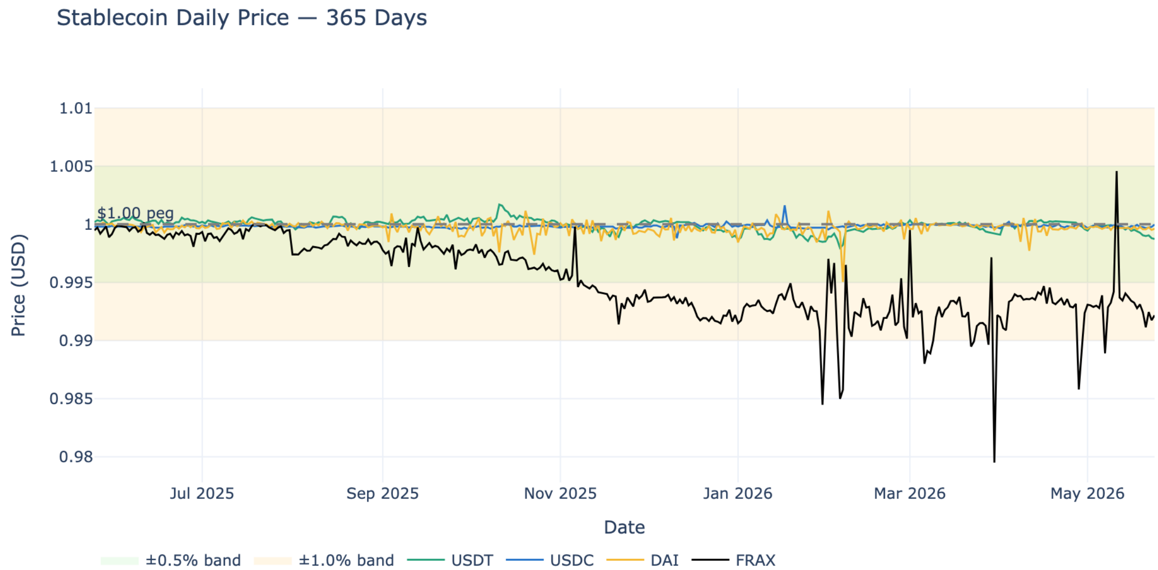

Fig. 1. Daily stablecoin prices around the $1 peg. The shaded bands represent $\pm 0.5\%$ and $\pm 1.0\%$ thresholds.

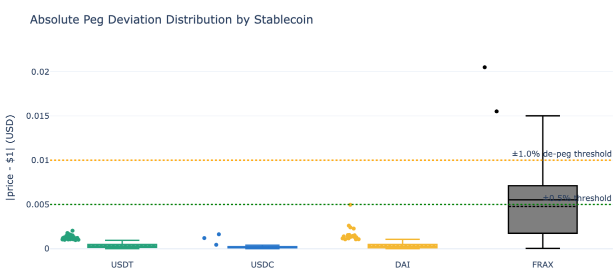

Fig. 2. Distribution of absolute peg deviation by stablecoin.

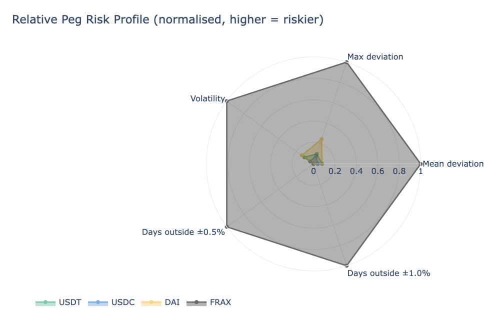

Fig. 4. Normalized relative risk profile. Higher values indicate higher relative risk on the selected metric.

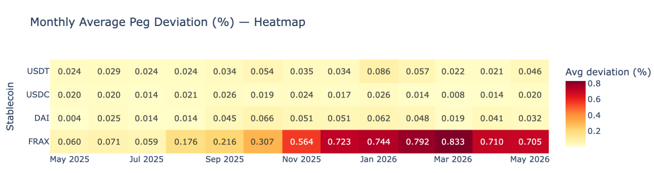

Fig. 3. Monthly average peg deviation. Darker cells indicate larger average deviation.

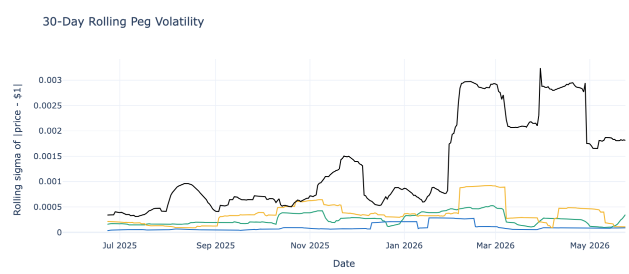

Fig. 5. Thirty-day rolling volatility of absolute peg deviation.

VIII. Risk Interpretation

The empirical findings motivates a mechanism-specific interpretation.

A. Fiat-Collateralized Stablecoins

The repository analysis indicates that USDT and USDC have the closest routine peg behavior. This helps them to be used as low noise settlement assets when they are used normally. But the volatility observed on a daily basis does not include their primary risk, which is a discontinuous shock that may come about due to failures by the issuer, the bank, the custodian, the reserve, the regulators, or the redemption channel [6], [7]. The peg breaks of fiat-backed designs can be fast and temporary, as shown in a recent example: the de-peg

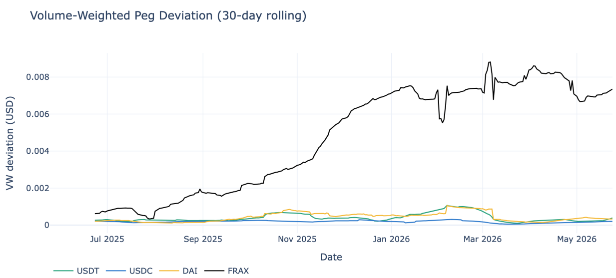

Fig. 6. Thirty-day volume-weighted peg deviation.

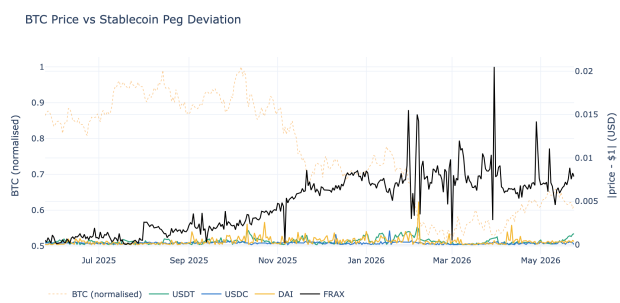

Fig. 7. BTC price behavior versus stablecoin peg deviation.

of USDC in March 2023, due to an issue with access to the reserve in Silicon Valley Bank.Such fast and short-lived peg breaks, like that of USDC in March 2023, following problems with access to the reserve in Silicon Valley Bank, illustrate this. This event is a contextual, and not necessarily time-in-between 365 days, unless the notebook is conducted over a period involving March 2023.

B. Crypto-Collateralized Stablecoins

The results showed a slightly larger baseline deviation for DAI. The cost is visibility: unlike a fully off-chain reserve system, collateral composition, vault parameters, liquidation process, and governance modifications are more observable in a system like MakerDao.The downside is visibility: in a fully off-chain reserve system, everything is less visible, including the composition of collaterals, vault parameters, liquidation conditions and governance changes, as in MakerDao. Although day-to-day price fluctuations may be larger, it may be easier for a risk manager to manage and hedge observable risk than opaque reserve risk.

C. Hybrid Stablecoins

The widest deviation behavior is seen in the repository findings with regards to FRAX. Reflexivity can occur with hybrid or fractional designs: When a portion of the stabilizing mechanism is crypto-native collateral or the value of the governance tokens, that backing can be compromised during market stress. The evidence from algorithmic and partially endogenous stablecoins suggests a vicious cycle of redemption pressure can set in when trust in the stabilizing asset breaks down, as seen in the runs of these stablecoins [3], [12]. This can result in a risk profile which is not suitable for a portfolio model with fiat-backed stablecoins (fiat stablecoins) (fractional reserve model) (frax$_{d}ocs$).

IX. Capital Allocation Implications

The primary allocation finding is that the exposure to stablecoins needs to be broken down by mechanism. You should never have a single undifferentiated 'stable coin' bucket in a DeFi yield book. Instead, you should start to differentiate fiat-backed from crypto-collateralized and hybrid designs, and each bucket's performance measured by deviation, liquidity, redemption and recovery. This aligns with policy-recommendations such as reserve quality, stablecoin redemption claims that are nigh-impossible to break, governance, operational resilience and risk management for stablecoin arrangements [7], [10].

Second, treat peg deviation as a part of real-time risk signal. Increasing the rolling deviation or the volume-weighted deviation may warrant reducing exposure, increasing the collateral haircut, increasing the collateral liquidity ratio or temporarily decreasing leverage in protocols backed by stablecoins. This is important because DeFi protocols can pass on shocks via collateral reuse, liquidations, and automated leverage [1], [8], [9].

Third, the statistical separation is important. It is reported that the significance of the Kruskal-Wallis test indicates that the four distributions of deviations should not be pooled at the p¡0.05 level without losing some of the crucial risk information. This helps to recognise exposures or capital charges on specific mechanisms, not to assume that all assets that are close to $1 are equally risky.

X. Limitations

Analysis is based on daily data and thus does not capture intraday de-pegging, rapid recovery processes and liquidity shocks that may occur during an exchange. CoinGecko's market data is aggregated across venues, which tells you the overall size of a market but will not necessarily be the same as a market's price that you can execute trades on in a specific DeFi pool or centralized exchange. The empirical window is also different when the notebook is rerun, due to the rolling 365-day sample. Future versions should store raw downloaded data, precise execution time, numeric results tables with charts for audibility.

XI. Conclusion

This paper presents an empirical report for a DeFi stablecoin peg-stability study. As seen in the results, the major stablecoins have varying amounts of tightness, and different forms of failure, to the peg of $1. Fiat-backed assets have low normal day deviation, but high off-chain tail risks; DAI has higher baseline noise, but higher on-chain observability; FRAX has high peg risk, and high reflexivity. The following differences are reflected in the deviation distribution, volatility in rolling deviations, volume-weighted deviations, and diagnostics of deviations during market stress. The takeaway for the practical application is obvious: smart allocations of DeFi capital should be based on the riskiness of a stablecoin, with peg deviation tracked dynamically, and where possible, all dollar-pegged assets not being treated as cash.

REPRODUCIBILITY

The analysis notebook is available as `stablecoin_peg_analysis.ipynb`. When the notebook will be runned again, it will download the latest data from CoinGecko and update the figures in the folder `charts/`. This paper's `.tex` source file is located in `paper/`.

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